

Transforming the CERN Collider: Constructing the Barrel Timing Layer



Caltech

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Abstract

This project involves constructing an **Assembly Test Stand** to help with the **proper installation** of the scintillator silicon photomultipliers (**Scintillator SiPMs**) to a tray that will be installed into the Barrel Timing Layer of the Compact Muon Solenoid in the **Large Hadron Collider at CERN**. The Assembly Test Stand successfully holds the tray, but the installation of the Scintillator SiPMs has not started, so its success is still undetermined.



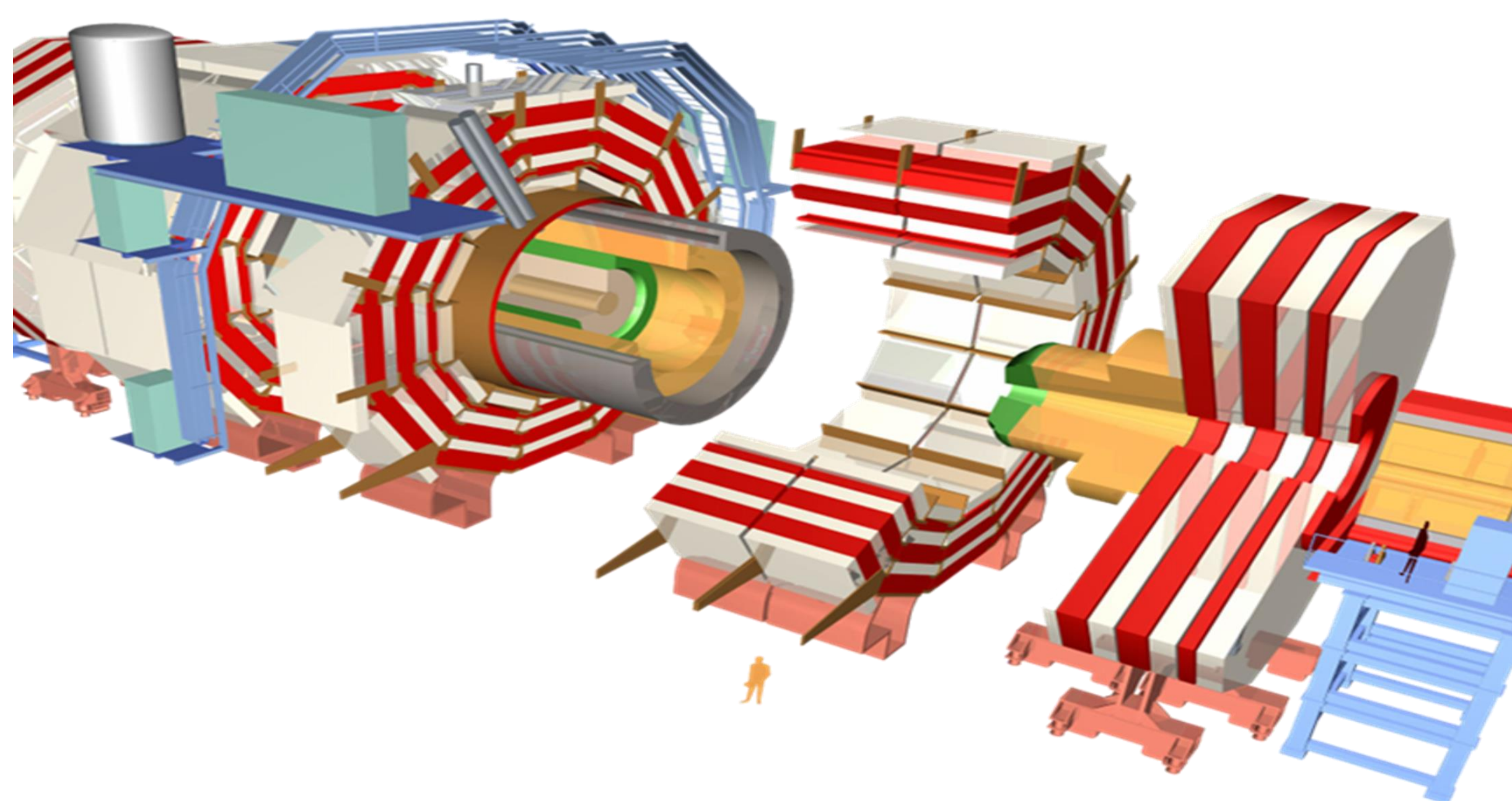
GALCIT

Method

- ✓Project engineers at CERN sent an assembly file of the Assembly Test Stand in SolidWorks
- ✓Determined which parts were standard and which needed custom manufacturing.
- ✓Ordered standard parts as well as the stock for custom parts from McMaster
- ✓Created drawing files for custom-made parts
- ✓Outsourced difficult custom-made to the GALCIT lab (use of CNC machine)
- ✓Manufactured by hand other custom parts
- ✓Built test stand from the bottom upwards
- ✓Discussed ambiguity in design and accommodated design changes during build

Background/Motivation

The Large Hadron Collider at CERN in Switzerland Geneva performs **high-energy particle collisions** to understand the standard model of particle physics. To continue its journey, each of the four detectors in the particle collider is being updated to handle collecting data from more particle collisions.



Inside the CMS detector is the new **Barrel Timing Layer (BTL)**. It detects the timing of individual photons entering/leaving the collisions and is made of trays that hold many scintillator silicon photomultipliers. **Caltech is responsible for building one of the trays.**

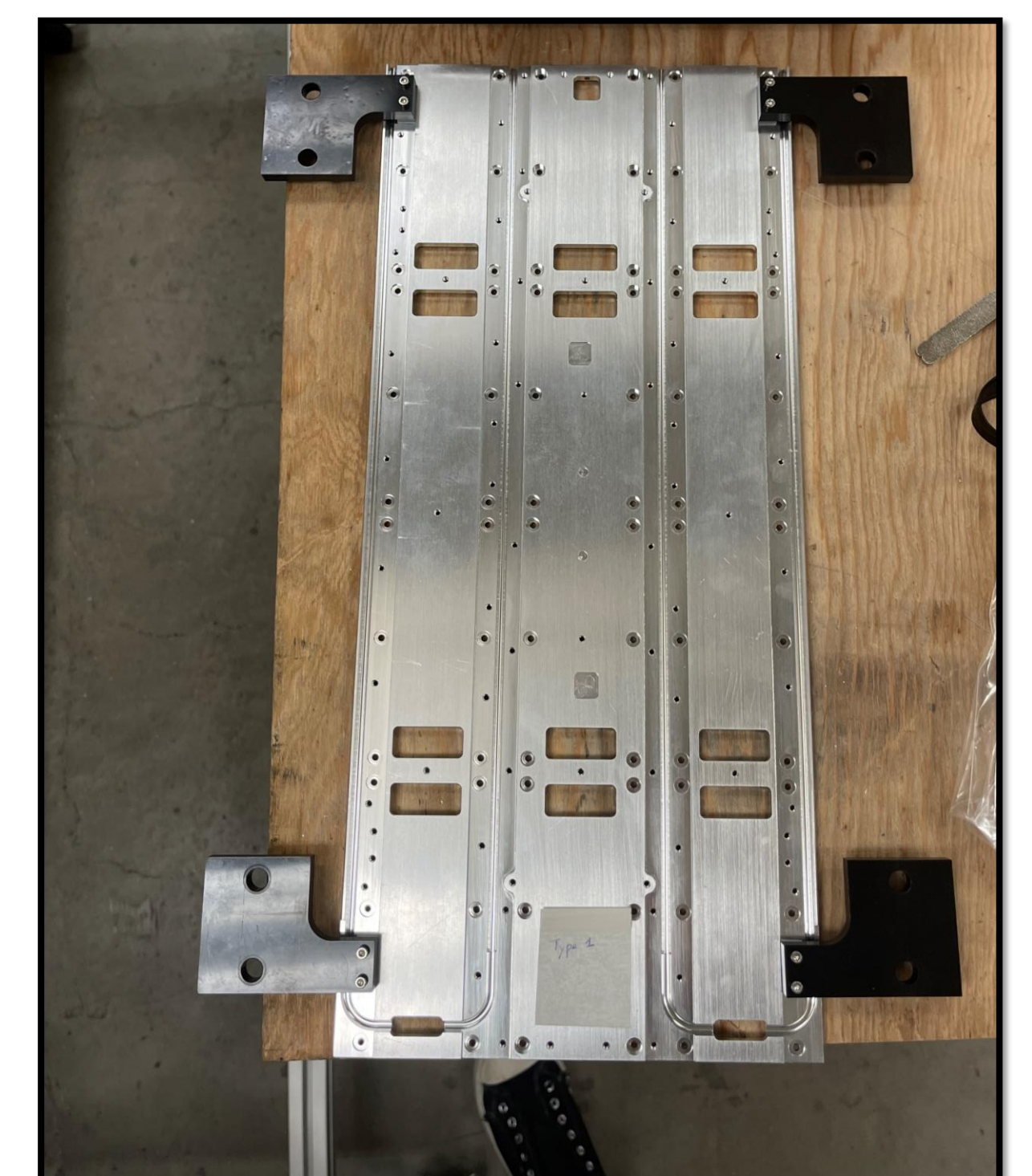
Results



Assembly Test Stand without tray



Assembly Test Stand rotated at 90 degrees



1 of 6 tray parts ready to be installed into the stand

Significance and Impact

- BTL construction is a major part of the **High Luminosity Update for CMS**
- The update allows for **more collisions** since each particle has better detection
- More observed collisions allow **higher chances of discovery**

Into the Future

- **Test** design of the Assembly Test Stand with tray
- Make necessary **upgrades** to the assembly test stand.
- Look into **constructing** more tools that allow **precise construction** of the BTL

